

PROJECT REQUIREMENT DOCUMENT

Custom Solar EPC CRM & ERP Platform for Greenvolt Energy Solutions LLP

Project Overview

We are a solar EPC company engaged in rooftop solar installations under PM Surya Ghar Yojana and MNRE subsidy programs. We require a web-based CRM and ERP platform that can manage the complete business workflow from lead generation to subsidy processing, loan management, installation, inventory, accounts, and after-sales service.

The platform should support multiple user roles including Admin, Dealer, Employees, Installation Teams, Loan Executives, MNRE Executives, and Customers.

The system should be scalable to support future expansion across Kerala and India.

MODULE 1: LEAD & CUSTOMER MANAGEMENT

Features:

Customer Database Management

Fields:

- **Customer Name**
- **Age**
- **Gender**
- **Mobile Number**
- **Alternate Number**
- **Email ID**
- **Installation Address**
- **District**
- **State**
- **Pincode**

Lead Source Tracking:

- **Facebook**

- **Website**
- **Dealer**
- **Referral**
- **Walking Customer**
- **Google Ads**
- **WhatsApp Campaign**

Lead Status Workflow:

- **New Lead**
- **Contacted**
- **Site Visit Scheduled**
- **Site Survey Completed**
- **Quotation Sent**
- **Negotiation**
- **Order Confirmed**
- **Installation Pending**
- **Installed**
- **Subsidy Pending**
- **Completed**

Document Storage:

- **Aadhaar**
- **PAN Card**
- **Electricity Bill**
- **Geotagged Photos**
- **Property Tax Receipt**
- **Land Tax Receipt**
- **Bank Passbook**
- **Customer Signature**
- **Other Supporting Documents**

Requirements:

- **Search Customers**
 - **Filter Customers**
 - **Lead Conversion Reports**
 - **Customer Activity Timeline**
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MODULE 2: DEALER PORTAL

Each dealer should receive separate login credentials.

Dealer Functions:**Create Customer:****Fields:**

- **Customer Name**
- **Age**
- **Gender**
- **Phone Number**
- **Email ID**
- **Installation Address**
- **Roof Type**
- **Project Size**
- **Inverter Brand**
- **Panel Brand**
- **Loan Required (Yes/No)**
- **Loan Amount**

Co-Applicant Details (if age above 65 years):

- **Name**
- **Mobile Number**
- **Relationship**

Dealer Capabilities:

- **Upload Customer Documents**
- **Request Quotation**
- **Confirm Orders**
- **Track Project Progress**
- **View Commission**
- **View Payment Status**
- **Download Documents**

Admin Functions:

- **Approve Dealers**
 - **Block Dealers**
 - **Assign Territory**
 - **Monitor Dealer Performance**
 - **Dealer Sales Reports**
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MODULE 3: SITE SURVEY MANAGEMENT

Survey Information:

- **Roof Type**
- **Roof Area**
- **GPS Coordinates**
- **Shadow Analysis**
- **Existing Electrical Load**
- **Sanctioned Load**
- **Phase Type (Single/Three Phase)**
- **Site Survey Photos**
- **Site Engineer Remarks**

Generate:

- **Survey Report PDF**
- **Recommended Plant Capacity**

- **Estimated Energy Generation**
 - **Preliminary BOM**
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MODULE 4: QUOTATION MANAGEMENT

Generate:

- **Quotation**
- **Proforma Invoice**
- **Tax Invoice**

Features:

- **Product Library**
- **Pricing Templates**
- **GST Calculation**
- **PDF Generation**
- **Digital Signature**
- **Approval Workflow**

Quotation Version Control

MODULE 5: PROJECT MANAGEMENT

Project Stages:

1. **Lead**
2. **Site Survey**
3. **Design**
4. **Quotation**
5. **Order Confirmation**
6. **Material Procurement**
7. **Installation**
8. **KSEB Application**
9. **Net Meter Approval**

- 10. Inspection**
- 11. Commissioning**
- 12. Subsidy Application**
- 13. Subsidy Received**
- 14. Project Completed**

Dashboard Should Show:

- **Delayed Projects**
 - **Upcoming Installations**
 - **Pending Approvals**
 - **Project Completion Rate**
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MODULE 6: MNRE & PM SURYA GHAR MANAGEMENT

Store:

- **Consumer ID**
- **Vendor ID**
- **Application ID**
- **Registration Number**

Track:

- **Registration Status**
- **Document Verification**
- **Inspection Status**
- **Subsidy Status**

Automated Alerts:

- **Missing Documents**
- **Inspection Due**
- **Subsidy Pending**
- **Application Rejected**

Document Storage:

- **Commissioning Report**
 - **Vendor Declaration**
 - **Installation Photos**
 - **Inspection Documents**
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MODULE 7: LOAN PROCESSING MODULE

Customer Loan Tracking

Fields:

- **Bank/NBFC**
- **Loan Amount**
- **Interest Rate**
- **EMI Amount**

Stages:

- **Applied**
- **Under Review**
- **Approved**
- **Sanctioned**
- **Disbursed**

Store:

- **Sanction Letter**
- **EMI Schedule**
- **Customer Agreements**

Reports:

- **Pending Loans**
 - **Approved Loans**
 - **Disbursed Loans**
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MODULE 8: INVENTORY & STOCK MANAGEMENT

Track Inventory:

- **Solar Panels**
- **Inverters**
- **Structure Materials**
- **ACDB**
- **DCDB**
- **Earthing Kits**
- **SPD**
- **Cables**
- **Connectors**
- **BOS Materials**

Functions:

- **Stock In**
- **Stock Out**
- **Warehouse Management**
- **Serial Number Tracking**
- **Low Stock Alerts**
- **Reorder Planning**

Multiple Warehouse Support

MODULE 9: PURCHASE MANAGEMENT

Features:

Purchase Request

Purchase Order

Goods Receipt Note (GRN)

Track:

- **Supplier**
- **Material Cost**
- **Quantity**

- **Delivery Status**

Supplier Database:

- **GST Number**
 - **Contact Details**
 - **Payment Terms**
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MODULE 10: INSTALLATION MANAGEMENT

Assign Teams:

- **Civil Team**
- **Fabrication Team**
- **Electrical Team**

Daily Updates:

- **Work Progress**
- **Installation Photos**
- **Materials Used**
- **Completion Percentage**

GPS-Based Attendance for Teams

Geo-tagged Completion Photos

MODULE 11: EMPLOYEE MANAGEMENT

Manage:

- **Staff Records**
- **Attendance**
- **Leave Management**
- **Payroll Integration**
- **Task Assignment**

Roles:

- **Admin**

- Sales Manager
 - Sales Executive
 - Dealer
 - Site Survey Engineer
 - Design Engineer
 - Procurement Manager
 - Store Manager
 - MNRE Executive
 - Loan Executive
 - Installation Manager
 - Installation Team
 - Accounts Executive
 - Service Engineer
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MODULE 12: FINANCE & ACCOUNTING

Integration:

- Zoho Books API

Track:

- Customer Payments
- Supplier Payments
- GST
- Receivables
- Payables

Project Profitability Calculation

Example:

Project Value = ₹2,00,000

Panel Cost = ₹95,340

BOS Cost = ₹20,000

Fabrication Cost = ₹12,000

Labour Cost = ₹12,000

CRM should automatically calculate:

- **Gross Profit**
- **Profit Percentage**
- **Net Margin**

Financial Dashboard:

- **Revenue**
 - **Monthly Sales**
 - **Outstanding Payments**
 - **GST Liability**
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MODULE 13: SERVICE & WARRANTY MANAGEMENT

Track:

- **5-Year Maintenance**
- **Panel Warranty**
- **Inverter Warranty**

Generate:

- **Service Tickets**
- **AMC Schedules**
- **Maintenance Reminders**

Customer Complaint Management

MODULE 14: CUSTOMER PORTAL

Customer Login

Customer Can:

- **View Project Status**
- **Track Subsidy Status**

- **Download Invoices**
 - **Download Warranty Certificates**
 - **Download Service Reports**
 - **Raise Service Requests**
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MODULE 15: DOCUMENT MANAGEMENT SYSTEM

Centralized Document Repository

Categories:

Customer Documents

Project Documents

MNRE Documents

Loan Documents

Supplier Documents

Features:

- **OCR Search**
 - **Version Control**
 - **Document Expiry Alerts**
 - **Download History**
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MODULE 16: NOTIFICATION SYSTEM

Notifications Through:

- **WhatsApp**
- **SMS**
- **Email**
- **In-App Notifications**

Examples:

- **Site Visit Reminder**
- **Loan Approval**
- **Material Arrival**

- Subsidy Approval
 - Warranty Expiry
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MODULE 17: WORKFLOW AUTOMATION

Examples:

Order Confirmed:

- Create Project
- Assign Site Engineer
- Create Procurement Request

Installation Completed:

- Create MNRE Task
- Generate Commissioning Documents

Subsidy Received:

- Mark Project Completed
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MODULE 18: VENDOR & SUPPLIER MANAGEMENT

Track:

- Panel Suppliers
- Inverter Suppliers
- BOS Suppliers

Store:

- Contact Information
- Price Lists
- GST Details

Monitor:

- Delivery Performance
 - Supplier Ratings
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MODULE 19: QUALITY CONTROL

Checklist Before Completion:

- **Earthing Verified**
- **ACDB Installed**
- **DCDB Installed**
- **SPD Installed**
- **Structure Inspection Completed**
- **Generation Test Passed**

Generate:

- **Quality Inspection Report**
 - **QC Certificate**
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MODULE 20: MANAGEMENT DASHBOARD

CEO Dashboard

Display:

- **Total Leads**
- **Active Projects**
- **Monthly Sales**
- **Revenue**
- **Gross Profit**
- **Net Profit**
- **Stock Value**
- **Dealer Performance**
- **Pending Subsidy Amount**
- **Pending Loan Amount**
- **Project Completion Rate**
- **Installation Performance**
- **Employee Performance**

Dashboard should support:

- **Graphs**
 - **Filters**
 - **Export to Excel/PDF**
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TECHNICAL REQUIREMENTS

Frontend:

- **React.js / Next.js**

Mobile App:

- **Flutter**

Backend:

- **Node.js (NestJS preferred)**

Database:

- **PostgreSQL**

File Storage:

- **AWS S3**

Authentication:

- **JWT + OTP Login**

Integrations:

- **Zoho Books**
- **WhatsApp API**
- **SMS Gateway**
- **Email Service**

Deployment:

- **AWS Cloud**

Scalability Requirement:

- **10,000+ Customers**
- **1,000+ Dealers**

- **Multi-Warehouse**
- **Multi-Branch**
- **Multi-State Expansion**

The system should be modular so future features such as AI quotation generation, AI subsidy eligibility checks, AI loan eligibility checks, and mobile applications can be added without major redesign.